Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/13/2005
PI Name:	Su, Ching-Hua Ph.D.		
Project Title:	Crystal Growth of ZnSe and Related Ternary Compound Semiconductors by Vapor Transport		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Polymers		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ching.h.su@nasa.gov	**Fax:**	FY 256-544-8762
PI Organization Type:	NASA CENTER	**Phone:**	256-544-7776
Organization Name:	NASA Marshall Space Flight Center		
PI Address 1:	Mail Code EM31		
PI Address 2:	Materials and Processing Laboratory		
PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	07/01/1996	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Kaukler , William	**Contact Phone:**	256-544-0693
Contact Email:	William.Kaukler@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Brebrick, R. (Marquette University) Berger, A. (Fisk University) Dudley, M. (State University of New York) Matyi, R. (University of Wisconsin) Ramachandran, N. (BAE Systems) Shih, H. (Texas Instruments)		
Grant/Contract No.:	Intramural		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ZnSe and related ternary compounds, such as ZnSeS and ZnSeTe, will be grown by a vapor transport technique and a real time in-situ non-invasive technique will be applied to monitor the partial pressures and growth surface morphology. The grown crystals will be characterized extensively to correlate the grown crystal properties with the growth conditions. 1. Crystal Growth a. Unseeded growth of ZnSe, ZnTe, ZnS, ZnSeS and ZnSeTe b. Unseeded growth of Cr-doped ZnSe c. Seeded growth of ZnSe, Cr-doped ZnSe, ZnSeS and ZnSeTe 2. In-situ Monitoring a. High temperature furnace with windows b. Automated UV/Vis/Near IR spectrophotometry to monitor the partial pressures along the path of the growth ampoule c. Optical interferometry and image analysis to study the surface morphology of the growing surface 3. Characterization of Grown Crystals 4. Theoretical Modeling a. One-dimensional diffusion limited theory b. Three-dimensional numerical simulation of mass transport and heat transfer. The investigation consists of extensive ground-based experimental and theoretical research efforts and concurrent flight experimentation. The objectives of the ground-based studies are to obtain the experimental data and conduct the analyses required to define the optimum growth parameters for the flight experiments, perfect various characterization techniques to establish the standard procedure for material characterization and quantitatively establish the characteristics of the crystals grown on Earth as a basis for subsequent comparative evaluations of the crystals grown in a low-gravity environment, and develop theoretical and analytical methods required for such evaluations. The materials to be investigated are ZnSe and related ternary semiconducting alloys (e.g., ZnSxSe1-x, ZnSe1-xTex, and Zn1-xCdxSe). These materials are useful for optoelectronic applications such as highly efficient light emitting diodes and low power threshold and high temperature lasers in the blue-green region of the visible spectrum. The recent demonstration of its optical bistable properties also makes ZnSe a possible candidate material for digital optical computers. Compositional non-uniformity, microstructural crystal defects (e.g., dislocations, small-angle grain boundaries, and second phase precipitates), and deviation from stoichiometry can seriously limit state-of-the-art device performance and future device applications. The reduction of gravity-driven convective fluid flows in a low-gravity environment is expected to be advantageous in minimizing these compositional variations and structural defects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No funding was received in FY04
Bibliography Type:	Description: (Last Updated: 02/18/2025)